

Assignment-Regression Algorithm

Problem Statement or Requirement:

A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same.

1.) Identify your problem statement.

Ans - We need to predict the insurance charges based on customer age, sex, bmi, children, smoker data as input.

2.) Tell basic info about the dataset (Total number of rows, columns).

Ans – 1338 rows and 6 columns are present in the dataset.

3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data).

Ans – There are 2 columns (sex, smoker) which is categorical column which needs to be converted into numerical. These 2 columns (sex, smoker) come under nominal data as the dataset of the columns cannot be compared with one another. These 2 columns (sex, smoker) can be converted into numerical value using One-hot encoding algorithm.

4.) Develop a good model with r^2_score . You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

Ans – On looking at the problem statement and dataset it is clearly evident that input and output is available and the output which we are going to predict is also numeric value. Hence this problem statement comes under **Machine Learning – Supervised Learning - Regression**

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5.) All the research values (r2_score of the models) should be documented.

Linear Regression: r_score – 0.78913454847886

SVM(Before_Data_Standardisation):-

s.no	Kernel	C	gamma	epsilon	r_score
1	linear	0.1	scale	0.1	-0.122129503
2	rbf	0.1	scale	0.1	-0.089596413
3	poly	0.1	scale	0.1	-0.086218277
4	sigmoid	0.1	scale	0.1	-0.089765308

SVM(After_Data_Standardisation):-

s.no	Kernel	C	gamma	epsilon	r_score
1	linear	0.1	scale	0.1	-0.080977867
2	rbf	0.1	scale	0.1	-0.089095494
3	poly	0.1	scale	0.1	-0.08832283
4	sigmoid	0.1	scale	0.1	-0.088290328

Decision tree: -

s.no	criterion	splitter	max_features	max_leaf_nodes	r_score
1	absolute_error	best	None	None	0.732088616
2	squared_error	best	None	None	0.655463818
3	poisson	best	None	None	0.67198352
4	absolute_error	random	None	None	0.717114892
5	absolute_error	random	1	2	-0.08740497
6	absolute_error	best	1	2	-0.083199537

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Random Forest: -

s.no	n_estimators	criterion	max_depth	max_features	r_score
1	100	absolute_error	None	None	0.856570154
2	100	squared_error	None	None	0.852464543
3	100	squared_error	10	None	0.86000644
4	100	poisson	10	None	0.859161695
5	100	poisson	10	1	0.858340306
6	100	squared_error	10	1	0.855288234

6.) Mention your final model, justify why u have chosen the same.

From the r2 score result random forest algorithm has the best r2 score of **0.86000644**(value near to 1). Hence chosen it has the final model.

AdaBoost Regressor:

estimator	n_estimators	learning_rate	loss	r_score
None	100	2	square	0.45764329
None	100	2	exponential	0.47034691
DecisionTreeRegressor(max_depth=4)	100	2	linear	0.6899524
DecisionTreeRegressor(max_depth=4)	100	1	linear	0.83492475
None	50	1	linear	0.86019772
None	100	1	linear	0.86019772
DecisionTreeRegressor(max_depth=3)	100	1	linear	0.86019772
None	100	2	linear	0.88095851

XGBoost Regressor:

n_estimators	learning_rate	max_depth	subsample	colsample_bytree	r_score
100	0.1	2	0.1	0.1	0.775642097
100	0.05	2	0.8	0.8	0.890605509
200	0.05	2	0.8	0.8	0.891699553
200	0.05	2	0.99	0.99	0.89242208

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LGBMRegressor :

<i>n_estimators</i>	<i>Learning_rate</i>	<i>max_depth</i>	<i>subsample</i>	<i>boosting_type</i>	<i>colsample_bytree</i>	<i>r_score</i>
200	0.5	2	0.99		0.99	0.8737
200	0.5	2	0.8		0.8	0.8780
200	0.5	2	0.8	dart	0.8	0.8907